Problem- Generate interrupt using “ebreak”

Program Requirements

1. Start execution in Machine mode.
2. Sets up a trap handler using mtvec
3. Executes an ebreak instruction to generate a breakpoint exception
4. In the trap handler:
    a. Reads the mcause register
    b. Checks whether the exception is caused by ebreak (mcause = 3)
    c. If true, updates the mepc register to skip the ebreak instruction
5. Returns from the trap handler using mret
6. Continues normal program execution after handling the exception

Observations Required

- ☐ mcause correctly identifies the exception
- ☐ Updating mepc skips ebreak
- ☐ Program continues successfully